8D Report: Door trim clips installed in the wrong orientation

FICTIONAL EXAMPLE. Written for a software demo. Every company, part and number in it is invented.

- Report number: 8D-2026-0417 (fictional)
- Supplier: Northfield Components GmbH (fictional)
- Part: Door trim retaining clip, drawing revision C
- Complaint raised: 3 March 2026
- Status: Closed

D1 — Team

A cross-functional team was formed on the day of the complaint: the supplier quality engineer (team lead), the assembly line supervisor, the process engineer responsible for work instructions, the design engineer for the clip, and a representative from the customer's incoming inspection.

D2 — Problem description

Customers reported a rattling noise from the front door trim. Inspection of returned vehicles found that the door trim retaining clips had been installed rotated by 180 degrees. In that orientation the clip seats only partly, and the trim panel can move against the door frame. 214 vehicles built between 10 and 21 February 2026 were affected. No safety function is involved.

D3 — Containment actions

Immediate actions were taken to stop more affected vehicles from reaching customers while the root cause was investigated:

- All finished vehicles from the affected build period were held and checked. Wrongly installed clips were reinstalled correctly.
- A 100% visual check of clip orientation was added at the end of the door line, with a reference photo at the station.

These actions stop the damage from spreading, but they do not fix the reason the clips were installed wrongly.

D4 — Root cause analysis

The team used the 5-Why method, asking why at each step until it reached a cause that, once fixed, would stop the problem from returning.

1. Why did customers hear a rattle? Because the clips were installed rotated by 180 degrees.
2. Why were the clips installed in the wrong orientation? Because the work instruction at the station showed the old orientation.
3. Why did the work instruction show the old orientation? Because it was not updated when the clip design changed from revision B to revision C, which turned the clip's locking tab around.
4. Why was it not updated after the design change? Because nobody was responsible for updating station work instructions when a part's drawing revision changes.
5. Why was nobody responsible? Because the design change process had no step that requires the affected work instructions to be identified and updated before the new revision reaches the line.

Root cause: the engineering change process does not require work instructions to be updated when a part design changes. The operators followed the instruction they were given; the instruction itself was out

of date.

The team noted that stopping at the second why would have led to retraining the operators. That would not have prevented the same problem with the next design change, on any part.

D5 — Permanent corrective actions

- The engineering change process now contains a mandatory step: for every drawing revision, the process engineer lists all affected work instructions and updates them before the change is released.
- A revision change cannot be released to production until that list is signed off.
- The clip received a small asymmetric rib, so that it physically cannot be pushed into the panel the wrong way round (mistake-proofing).

D6 — Implementation and verification

The corrective actions were implemented on 24 March 2026. During the following six weeks, zero wrongly installed clips were found, and the end-of-line 100% check was then removed. The effectiveness was verified with the customer's incoming inspection data.

D7 — Prevent recurrence

- The same change-process step was rolled out to all part families at the plant, not only door clips.
- A check of 40 other recent design changes found two more outdated work instructions, which were corrected before they caused defects.
- The lesson was added to the training for new process engineers.

D8 — Closure

The team presented its results to plant management on 15 May 2026, and the complaint was closed. The team was recognised for finding a cause in the process, rather than stopping at the operators.